

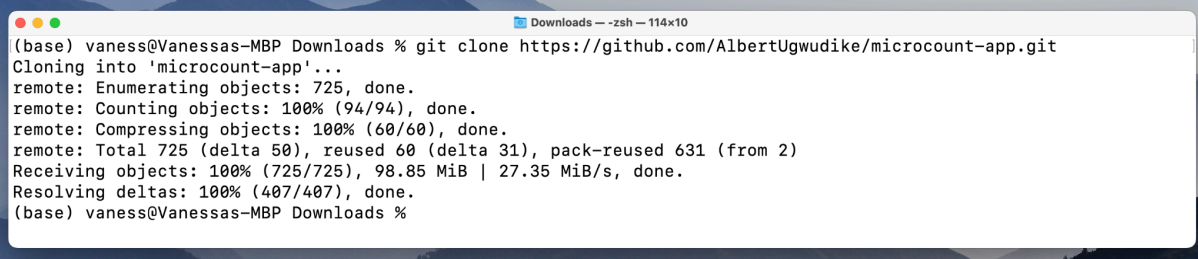
Microcount Protocol

Requirements

- MATLAB >= R2022b
- Bioformats Package (MATLAB Add-On manager)
- Parallel Computing Toolbox (MATLAB Add-On manager)
- Image Processing Toolbox (MATLAB Add-On manager)
- Java >= 21.0.2 (<https://www.oracle.com/uk/java/technologies/downloads/>)

Installation

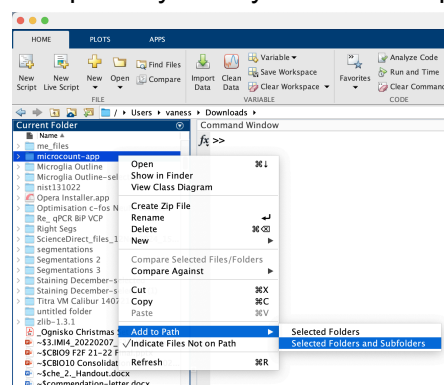
- Clone the repository
- Use command: 'git clone <https://github.com/AlbertUgwudike/microcount-app.git>'



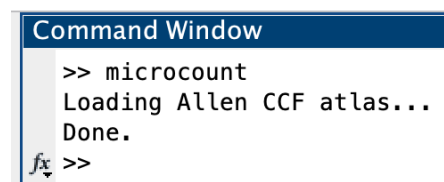
```
(base) vaness@Vanessas-MBP Downloads % git clone https://github.com/AlbertUgwudike/microcount-app.git
Cloning into 'microcount-app'...
remote: Enumerating objects: 725, done.
remote: Counting objects: 100% (94/94), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 725 (delta 50), reused 60 (delta 31), pack-reused 631 (from 2)
Receiving objects: 100% (725/725), 98.85 MiB | 27.35 MiB/s, done.
Resolving deltas: 100% (407/407), done.
(base) vaness@Vanessas-MBP Downloads %
```

Usage

- Open MATLAB
- Navigate to the downloaded repository
- Ensure the downloaded repository is on your MATLAB path



- Run the command 'microcount'



```
Command Window
>> microcount
Loading Allen CCF atlas...
Done.
>>
```

- The following window should appear:



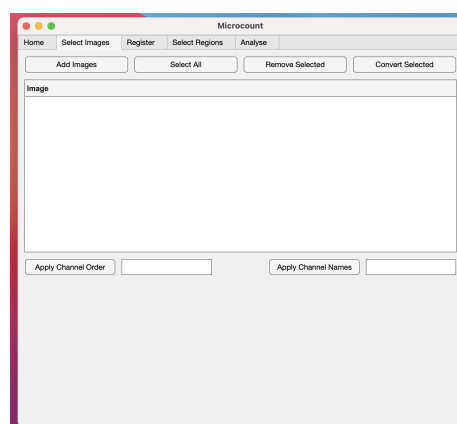
- On this tab you are able to create a new workspace or load an existing one.

You can think of a workspace as a particular set of related images, settings and results. Users may use different workspaces for different projects.

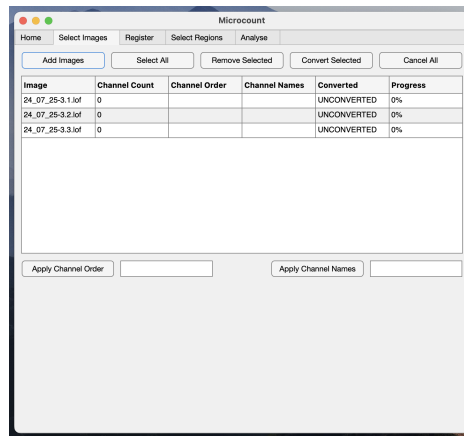
- Click 'Create Workspace' to open a window enabling you to name and create a new workspace. **Skip this step if you already have one created.**
- Then, click 'Load Workspace' to select the created workspace folder. The full path to this folder should appear beneath the buttons
- Once a workspace is loaded you may now navigate to the other sections of the pipeline

Note: Tabs are arranged in a sequential manner. If you have not selected/converted images, there will be no images to register. If there are no images registered, there will be no images for region selection etc..

- Click the 'Select Images' Tab. You will be presented with the following:



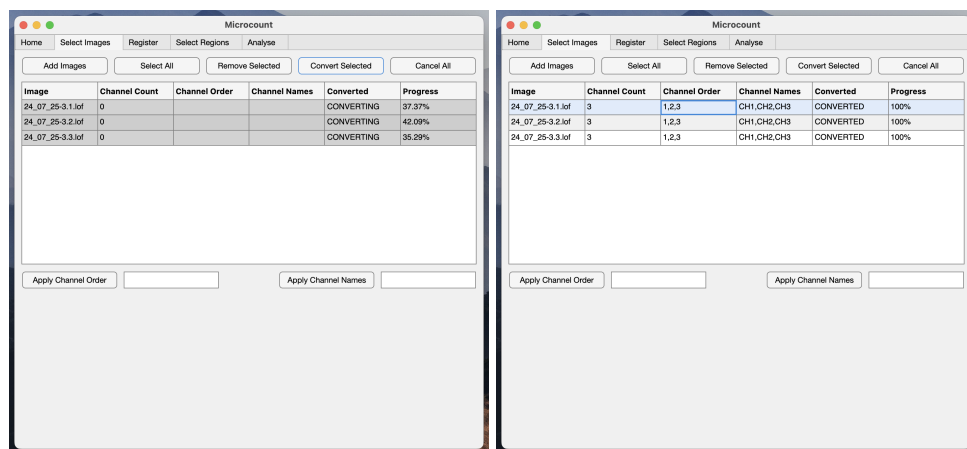
On the 'Select Images' Tab you are able to select images for further processing, additionally the channel order and specific channel names can be specified on this page as well.



When images are first added you'll notice that 'Channel Count', 'Channel Order' and 'Channel Names' are blank. These will be populated once images are converted into a more general format.

Conversion can be achieved by highlighting the desired images and clicking the 'Convert Selected' button. For large images, this can take a couple minutes. Progress of the conversion is displayed in the 'Progress Column'

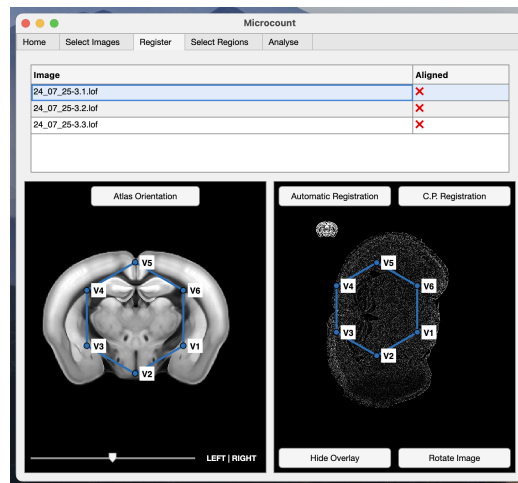
**** I'm currently working on improving the memory efficiency of conversion, I recommend converting a maximum of five images at a time to avoid memory issues ****



Microcount will utilise Channel 1 for registration, Channel 2 for cell segmentation (IBA1) and Channel 3 as the co-marker channel (CD68). Users are able to change the channel order for converted images. Select the images you want to configure, enter the new channel order as a comma separated list of digits (e.g. '2,1,3') and click 'Apply Channel Order'. Use a similar approach to assign unique channel names if desired.

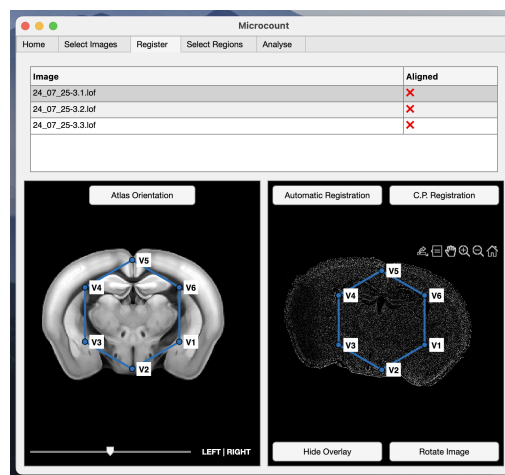
The table can also be modified directly on a per-image basis.

- Once one or more images have been converted, navigate to the 'Register Tab'
- On the register tab, images available to register will be shown in the table. Click on one of them to display its thumbnail:



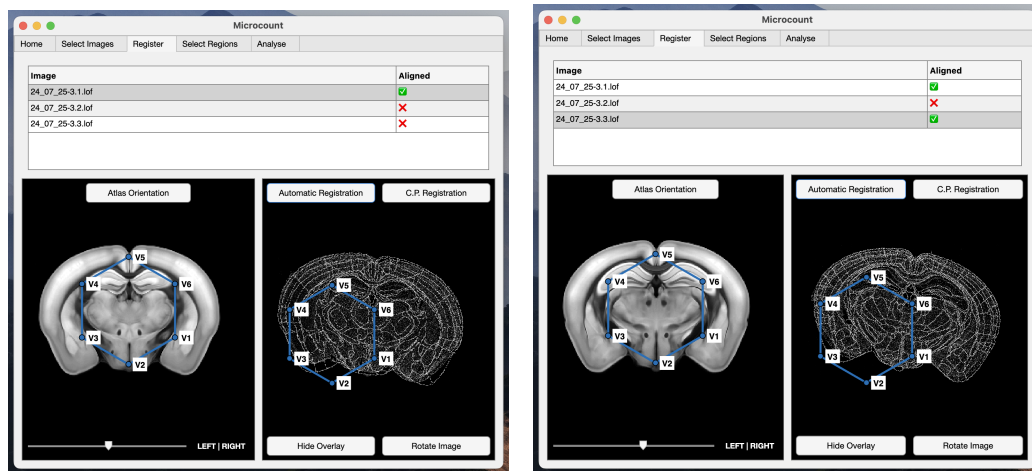
On the register tab there are various tools to assist with alignment, you are able to rotate the input image and also toggle the orientation of the atlas between axial, sagittal and coronal.

- Use the 'Rotate Image' button to rotate the input image to the correct position.
- Use the Atlas slider to choose the correct atlas depth

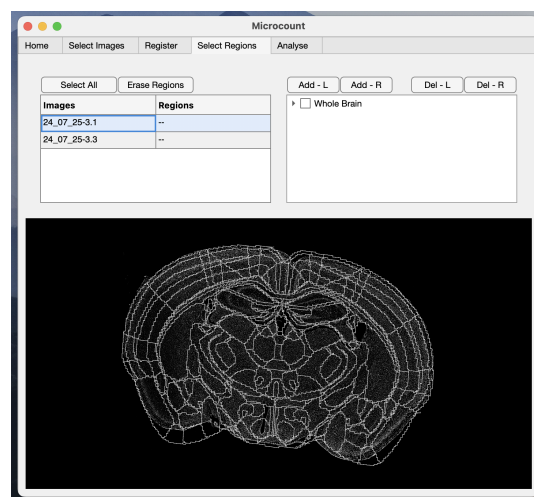


Now that both our brain section and atlas section are in similar positions we can now attempt to register. There are two strategies for registration that can be used 'Automatic Registration' tries to align the images based on intensity information. 'C. P. Registration' attempts to align images using corresponding control points (depicted as the vertices of a blue hexagon here)

- Use 'Automatic Registration' to register the image and atlas slice

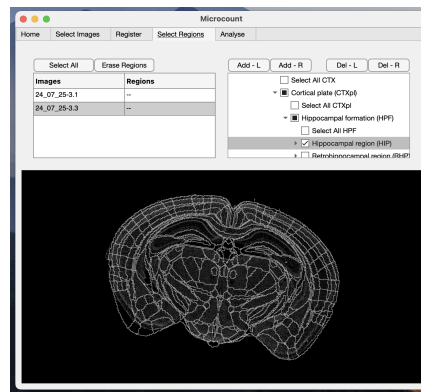


- Now that images have been registered we you may go to the Region Selection Tab
- On this tab select an image and you will be presented with the following view



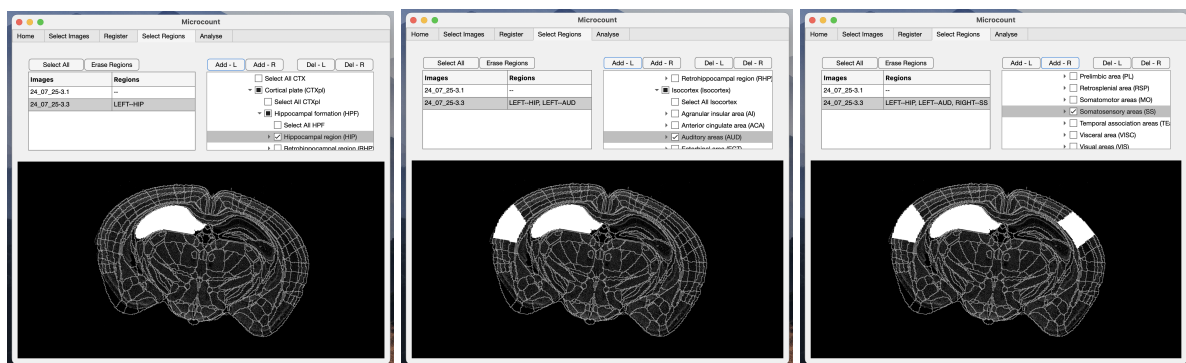
On the select regions tab, users are able to select anatomical regions of interest within each registered image for further processing. The selection of regions is achieved by navigating and selecting components of the Region Structure Tree.

- Select your anatomical regions of interest in the region selector tree



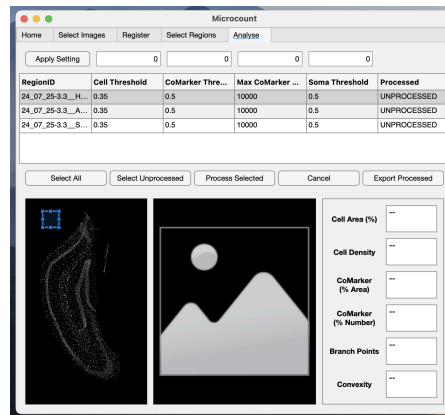
- Once selected you can add these regions to either the left or the right side of the registered mask by clicking the respective button. Regions may be removed from the left or right of selected images in a similar fashion!

NOTE: Removing regions that have already been analysed (later on) in this step effectively deletes the analysis results for that region.



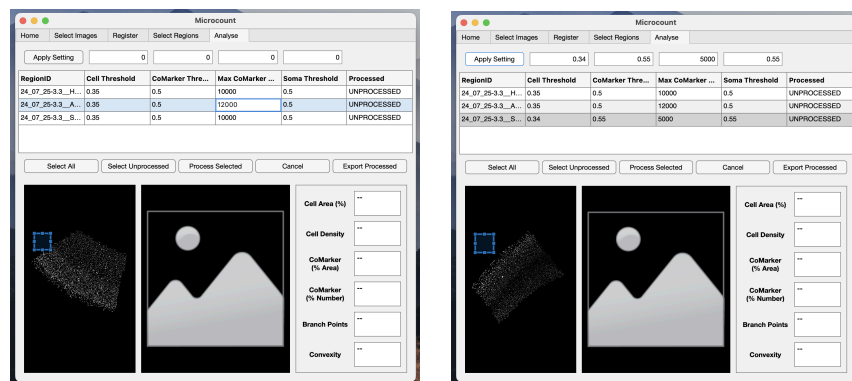
Note: If a region is not present in the registered mask, it will not be added for analyses (clicking either of the 'add' buttons will do nothing in this case). Furthermore, if the registered slice is in the sagittal plane then only the left OR right buttons will appear to do anything (depending on the slice depth)

- Now that regions have been selected, you may navigate to the 'Analyse Tab'
- Click on one of the regions and you will observe the following:

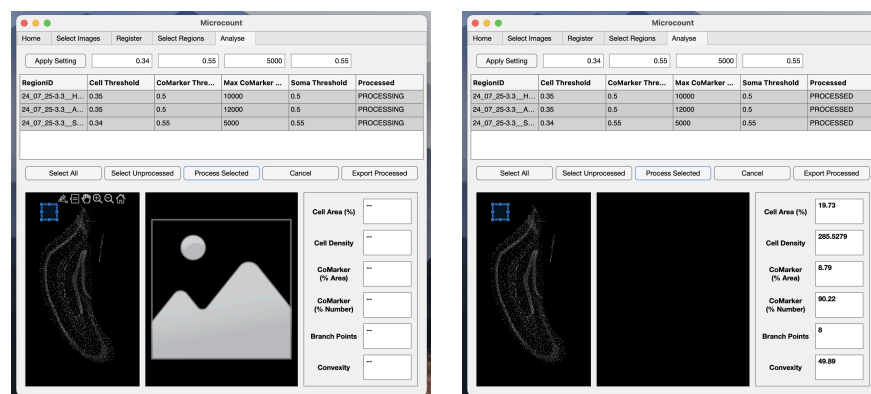


In the 'Analyse Tab' you are able to set algorithmic parameters (to individual images by entering into the table, or to multiple images at once using the top fields and 'apply setting' button.

- Adjust the parameters passed to microcount on a per-region or whole-dataset basis as required



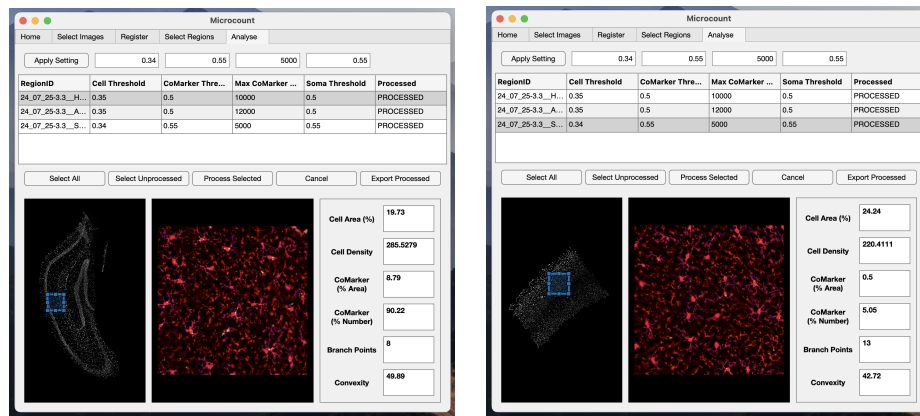
- Highlight the images you would like to process and click 'Process Selected' to begin analysis.



Note: For larger images this step can take quite some time, we found that for ~5Gb images, analyses would complete in a few minutes.

Note: The two main computationally intensive tasks (image conversion/downsampling and image processing) have been implemented to run in parallel in the background. Once a region is marked as complete you may proceed whilst other images are processing.

- Once a region has been processed, select it for viewing.
- Move the region-viewer window (blue-rectangle) to the portion of the downsampled image you would like to view.



Note: The region-viewer was implemented in this way due to the performance-cost of rendering full-size images within the app. If you would like to see the full image in your FOV, you can reshape the region-view window to occupy the full extent of the downsampled image.

- Adjust parameters and re-process as you deem necessary
- Once satisfied with the analysis output, you may click the 'Export Processed' button to export the data from all processed regions
- The exported data should look similar to this:

example

file_name	mask_file_name	region_ID	cell_threshold	max_comarker_size	comarker_threshold	cel_density	percentage_cell_area	percentage_comarker_area	percentage_comarker_num	average_convexity	average_soma_size	average_branch_count
/Volumes/Mort	/Volumes/MorseDrive	24_07_25-3.2	0.35	10000	0.5	285.5279	19.73	8.79	90.22	49.89	55.0967	8
/Volumes/Mort	/Volumes/MorseDrive	24_07_25-3.2	0.35	12000	0.5	282.9163	17.51	1.43	36.59	49.18	60.8645	6
/Volumes/Mort	/Volumes/MorseDrive	24_07_25-3.2	0.34	5000	0.55	220.4111	24.24	0.5	5.05	42.72	47.3869	13